



**The Specialists
for Pectin**



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THE SPECIALISTS FOR PECTIN – *a Product of Nature*



The Corporate Group Herbstreith & Fox is world-wide a byword for makers of high-quality pectins used in the food and non-food industry for more than 75 years.

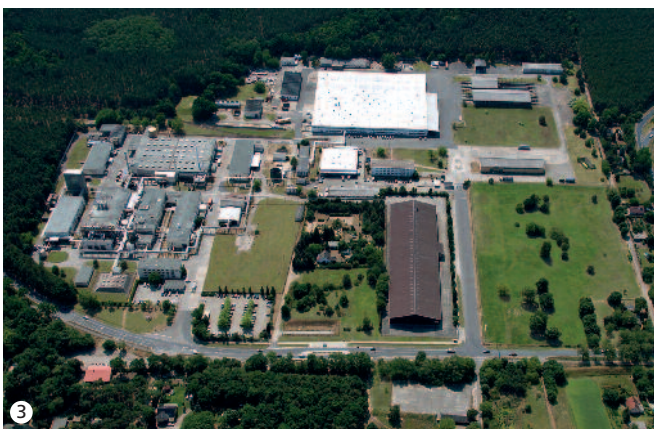
Decades of experience with perspectives in research and development, in production and sales are certainly the foundation for the present success in the world market.

Tradition and experience alone will not be enough to be successful in today's markets. Only with a modern management, promoting innovative thinking and not wishing to be merely copycats but pioneers, listening to the market demands, will it be possible to transform ideas into promising action. Herbstreith & Fox has always consistently followed this corporate philosophy.

In this way, convincing innovative products at the highest quality level were created in the most advanced production processes.

Ideas can only be successfully transformed by modern management, promoting innovative thinking.

The Corporate Group Herbstreith & Fox provides users of pectins with a sophisticated high-quality product range of all types of pectins, which meet the most demanding requirements across a broad spectrum of applications.



① The pectin works at Neuenbürg are cradle and headquarters of the whole corporate group.

② Rotenbach: Raw materials and finished goods warehouse with an area of 15,000 m².

③ The production site at Werder was acquired in 1990. It is at the same time headquarters of Herbafood Ingredients GmbH and agro Food Solution GmbH.

TAKEN *from Nature*

Pectin is a natural substance which is found in many fruits and vegetables. As structural element in the growth generating tissue and main component of the middle lamella in plants it provides cohesion and stability in tissues and cells.

Pectin as a name is derived from the Greek word “pectos”, which means gelatinated, or solidified.

Pectin is a relatively new ingredient though the scientist Vauquelin discovered its existence in fruit juices some 200 years ago.

At that time, however, in 1790, the year of birth for pectin, it was not yet called by that name. The name was first used in 1824, when Braconnot continued the research started by Vauquelin. He called the gel-forming substance pectic acid.

Smolenski in 1924 was the first to assume a substance from polymeric galacturonic acid in pectin. In 1930, Meyer and Mark discovered the chain formation of the pectin molecule. And Schneider and Bock in 1937 established the formula.

The practical potential was recognized only at the beginning of the 20th century. The excellent jellifying properties of pectin were employed in the production of foodstuff, which marked the beginning of the large-scale production of pectin.

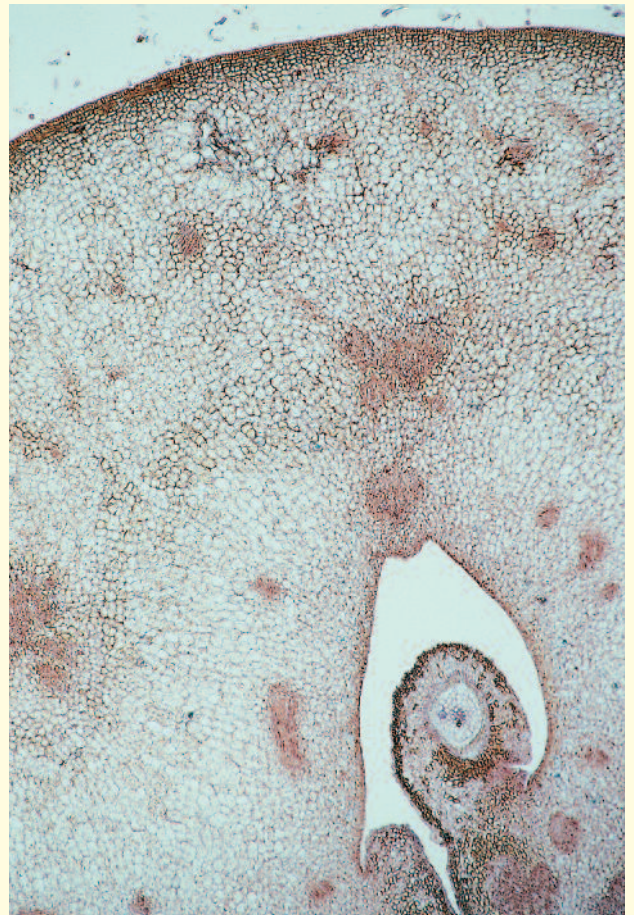


Fig. 1: Photomicrography of a tissue of an apple slice

The founder of our company, Hermann Herbstreith, discovered decades ago the usefulness and multivariied application potential of pomace, a by-product from the fruit juice manufacture. From then on, the undervalued pomace served as valuable raw material in the production of apple pectin.

Chemically speaking, pectin is a macromolecular compound belonging to the heteropolysaccharides. The main component is polygalacturonic acid, which is partially esterified with methanol.

Pectin is recognised by consumers and manufacturers alike as clean label.

Pectin is produced from fruit or vegetable raw material with a high natural pectin content, such as apple pomace, citrus peels and beet chips, in a technologically sophisticated process.

The different raw materials will yield various amounts of pectin. These pectin extracts are now specifically processed in various processes to achieve a number of different types of pectin with specific properties.

Choosing a pectin type for customers is dependant on which properties need to be present in the final product.

Pectin is today an indispensable component in many products, mainly in the food sector, but also in the non-food sector, since it is a universally applicable, naturally gelling, thickening and stabilising substance.

***Pectin content of
different raw materials***

apple pomace	10 – 15 %
sugar beet chips	10 – 20 %
citrus peels	20 – 35 %



Fig. 2: Raw materials, which are a source of pectin

OUR PRODUCT VARIETY *is the answer to Market Requirements*

Herbstreith & Fox are widely recognised around the world as the most technical pectin producer on the market. For over 75 years we have focused on perfecting a range of premium quality pectins. Our pectin expert team are pushing the boundaries of pectin every day for new applications.

The apple has always been of outstanding importance for the production of pectin.

Raw materials for the manufacturing of our high, medium and low methylester Classic Apple and Citrus Pectins are **apple pomace** and **citrus peels**.

High gel strength and excellent thickening properties of Classic Pectins provide products with an extraordinary texture achieved with a small amount of pectin. Classic Pectins have a neutral taste and enhance the existing natural flavour of the processed fruit.

From our expertise in the production of **Classic Apple** and **Citrus Pectins**, we also produce high-quality **Classic Grapefruit Pectins** and **Betapec Beet Pectins**.

Thus the Herbstreith & Fox product range supplies today the full scope of commercially available pectins for all areas of application, such as:

- Classic Apple Pectins
- Classic Citrus Pectins
- Classic Grapefruit Pectins
- Betapec Beet Pectins

Aside from Classic Pectins, we also supply for special applications and for achieving specific properties tailored **Combi Pectins**, made by blending various classic raw materials.

The group of **Amid Pectins** presents a technologically reasonable supplement and alternative to our Classic Pectins. The specific properties of Amid Pectins lead to different gel structures and will provide technological benefits in certain areas of application.

Such a comprehensive product range covers all texture requirements.

For direct blending into the product batch and thus easier and quicker handling, we offer users **Instant Pectins**, which permit a lump-free dispersion.

And last but not least, we offer **Pectin à la carte**, specially designed to meet your requirements and specifications.

Our extensive, highly technical range is also Kosher and Halal certified.

The importance of pectin for the quality of the finished product is very high and the Corporate Group Herbstreith & Fox, the pectin specialists, will meet all requirements existing in practical work with a comprehensive product range.



ALL IMPORTANT: DEGREE OF ESTERIFICATION and Gelling Properties

Pectins are divided into three groups on the basis of their different gelling properties.

High Ester Pectins (HM-Pectins)

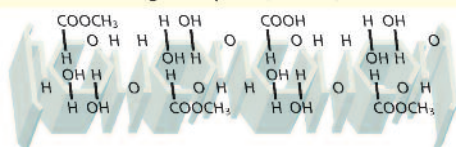
These pectins have usually a more than 50% share of esterified polygalacturonic acid units (DE°), thus practically no reaction with calcium ions occurs. The gel strength depends, among others, on acid content, type of pectin, concentration and soluble solids content, which generally has to be more than 55%.

The degree of esterification correlates with the gel setting rate and gel texture under otherwise similar conditions. Very high esterified pectins jellify quicker resp. at higher temperatures and form more elastic and brittle gel textures than less esterified pectins.

Your texture requirements determine the choice of the suitable pectin.

This accurate correlation requires a very homogeneous inter- and intramolecular carboxyl group distribution, as is especially given in Classic Apple Pectins from Herbstreith & Fox. Due to a blockwise distribution of carboxyl groups Classic Citrus Pectins with the same degree of esterification will form gels with a slightly higher setting temperature and a more elastic-brittle texture than Classic Apple Pectins. The blockwise carboxyl groups distribution of high methylester pectins additionally provides advantages regarding protein stabilisation in acidified milk drinks.

High ester pectin (75% DE°)



Low Ester Pectins (LM-Pectins)

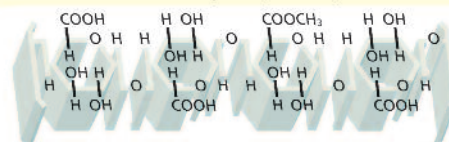
Pectins with less than 50% share of esterified polygalacturonic acid units (DE°) can jellify with

calcium ions. LM pectins thus do not only form gels with sugar and acids, but at less soluble solids mainly with calcium ions. The resulting gel strength is determined by pectin concentration, type of pectin, soluble solids content, pH range and the concentration of buffer salts and calcium ions. A well matched balance between pectin and calcium concentration will lead to an optimal texture.

Exceeding the calcium optimum will produce a brittle gel with tendency towards syneresis (loss of water from the gel) or, in the end, to the formation of calcium-pectinate, the insoluble calcium salt of pectin.

Since gel setting with LM pectins is also possible with a low soluble solids content and/or at a high pH-value, this opens up numerous application possibilities in dietetic and dairy products.

Low ester pectin (25% DE°)

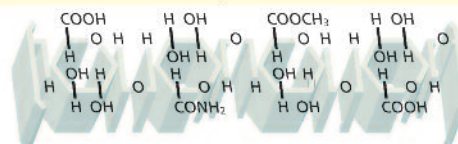


Amidated Pectins (HM- and LM-amidated Pectins)

In case of amidated pectins ammonia instead of acids is used for deesterification and with that part of the ester groups is replaced by amide groups. This process modifies the gelling properties in comparison to acid deesterified pectins.

LM amidated pectins, just like non-amidated pectins, require calcium ions for gelling. They will already set sufficiently with only minor calcium amounts present. Furthermore, the gel setting temperature of amidated pectins is less influenced by the calcium dosage.

Low ester amidated pectin (25% DE°, 25% DA°)



SOPHISTICATED METHODS FOR STANDARDISATION *Guarantee Reproducible Results*

Since pectin is extracted from natural vegetable raw materials, its properties may differ depending on the quality of the raw material.

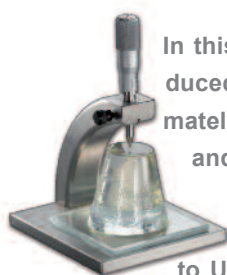
We, as pectin specialists, must supply users with products for large-scale industrial production that guarantee reliable and reproducible results on the basis of a standardised pectin grade.

Standardised pectin quality guarantees reproducible product properties.

Depending on the use of pectin, different methods of standardisation are being employed. Three essential methods:

USA-Sag-Methode

The standardisation of high methylester pectins to constant grades is internationally done by way of a **Ridgelimeter** according to the “USA-Sag Method”.



In this method, a sugar-water gel is produced at 65 % TSS and a pH of approximately 2.2. After a defined cooling period and cooling temperature, the percentage of sagging under its specific weight is measured and converted to USA-Sag. A frequent standard is 150 °USA-Sag. The disadvantage of the method is that the low pH range of the test gel does not recommend itself for practical work and it does not provide any information on the gel texture. The simple handling and good reproducibility are the advantages of this method.

Measuring breaking strength with the Pektinometer Mark IV

The “inner strength” of a gel is called “breaking strength”. It correlates more with the sensory perception of stability than the USA-Sag value.

Thus pectins often are assessed also for their breaking strength. To determine the breaking strength, the internal gel stability of a standard gel (65% TSS, pH 3.0) is measured by way of the **Herbstreith Pektinometer** by pulling a central shear insert out of the gel at constant velocity. The force necessary for this purpose is measured by a strain gauge.



The force variation is additionally recorded as a force-displacement diagram on the display providing information on the gel texture.

Setting temperature and setting time

Standardisation of setting temperature and setting time is done by using high-quality **Rheometers**. These measurements are supplemented by texture analysis. Thus the dynamic Weissenberg index provides values which correlate well with sensory impressions.

The thickening properties of pectins such as formation of a yield point or the viscosity are also determined rheometrically.

In penetration measurements with the **Texture Analyser**, the force of resistance is measured which acts on a test plunger penetrating a gel over a defined path or disrupting into it, respectively.



PACKAGED TO *Customer Specifications*

Once the pectin has passed through all stages of the production process and quality control, it is suitably packaged to start its journey to various industries.

Specific customer requirements will be considered during filling and package composition.

Our pectin is ordinarily filled into 25 kg cardboard boxes with an inside polyethylene bag. Specific customer requirements regarding filling and composition of packages may be considered. This high standard of packaging means pectin may be stored cool and dry for up to 18 months.

A worldwide distribution network assures reliable delivery of the products to our international customers.

Standard Packaging Sizes*:	
Boxes	25 kg
Big Bag	500 kg
	800 kg

* special sizes upon request



CERTIFIED *Product Quality*

The reliably high quality of our valuable pectins has decisively contributed to our success in the world market. Only with a continuous system of in-process tests through all stages of production, from suppliers of raw materials to the finished product, we are able to guarantee a uniformly high quality level of our products.

After many successful years of quality management according to DIN EN ISO 9001, the internationally accepted food safety management system FSSC 22000 was introduced with the initial certification in October 2014.

With the awarded certificate we could provide evidence for a functional management system in the food safety sector corresponding to the strict requirements of the international

FSSC 22000 norms which are completely approved by the Global Food Safety Initiative (GFSI).

The compliance of requirements of the QS-System for animal feed has been successfully and continuously proven since 2013 within the framework of system audits as well as spot audits.

The Herbstreith & Fox corporate group does justice to the responsibility to produce high grade and safe food ingredients and feed materials by extensive quality assurance measures.

A uniformly high level of quality is guaranteed.



ENERGY MANAGEMENT *Energy savings and Energy efficiency*

As an energy-intensive company we have already been paying attention to an efficient use of energy for years. We contribute to the protection of the resources and we are realising an optimal production process under economic and ecological aspects. In this process our employees make a significant contribution to the continuous improvement of our energy efficiency. With our energy politics we commit ourselves to comply with all relevant legal requirements of the energy law and to guarantee a reliable, resource saving and economical provision and usage of energy in our company by continuous improvements.

With the awarded certificate we have already provided evidence that our Energy Management System complies with the international standard DIN EN ISO 50001. With a multitude of measures we have already succeeded to make our production process more energy efficient. With the continuous energetic optimization of our production, we want to contribute to a higher product sustainability, lower CO₂-emissions and therefore to climate protection.



REGULATIONS FOR

Purity Requirements of Pectins

International Specification	EU E 440 (i) Pectin	EU E 440 (ii) amidated Pectin	FAO/WHO JECFA Pectins	FDA/FCC Pectins	USP Pectin
1. Loss on drying	max. 12.0 %	max. 12.0 %	max. 12.0 %	max. 12.0 %	max. 10.0 %
2. Acid-insoluble ash (in approx. 3N HCl)	max. 1.0 %	max. 1.0 %	max. 1.0 %	max. 1.0 %	–
3. Total insolubles	max. 3.0 %	max. 3.0 %	max. 3.0 %	max. 3.0 %	–
4. Sodium methyl sulfate	–	–	–	max. 0.1 %	–
5. Free methyl, ethyl, isopropyl alcohol	max. 1.0 %	max. 1.0 %	max. 1.0 %	max. 1.0 %	max. 1.0 %
	on the anhydrous basis				
6. Sulphur dioxide	max. 50 ppm	max. 50 ppm	max. 50 ppm	max. 50 ppm	max. 50 ppm
	on the anhydrous basis				
7. Nitrogen content (pectins) (after washing with acid and ethanol)	max. 1.0 %	–	max. 2.5 %	–	–
8. Nitrogen content (amidated pectins) (after washing with acid and ethanol)	–	max. 2.5.0 %	max. 2.5 %	–	–
9. Galacturonic acid (on ash-free and anhydrous basis)	min. 65.0 %	min. 65.0 %	min. 65.0 %	min. 65.0 %	min. 74.0 %
10. Degree of amidation (amidated pectins)	–	max. 25 %	max. 25.0 %	max. 25.0 %	–
11. Sugar and organic acids	–	–	–	–	max. 16.0 %
12. Arsenic	max. 3 ppm	max. 3 ppm	–	–	max. 3 ppm
13. Lead	max. 5 ppm	max. 5 ppm	max. 2 ppm 0.5 mg/kg for use in infant formula	max. 5 ppm	max. 5 ppm
14. Cadmium	max. 1 ppm	max. 1 ppm	–	–	–
15. Mercury	max. 1 ppm	max. 1 ppm	–	–	–
16. Total aerobic microbial count	–	–	–	–	max. 1.000 CFU/g
Yeasts and molds	–	–	–	–	max. 100 CFU/g
17. Pathogenous germs	according to general food regulations				
18. Pesticides	according to general food regulations				

– = no specifications

EU = European Union
FAO/WHO = Food & Agriculture Organisation/World Health Org.
JECFA = Joint Expert Committee on Food Additives

USP = United States Pharmacopeia
FDA = Food and Drug Administration
FCC = Food Chemical Codex

H&F-WORLDWIDE

Wherever you are – we are close to you

- Algeria
- Australia
- Austria
- Baltic States
- Belarus
- Belgium
- Canada
- China
- Czech Republic
- Denmark
- Egypt
- Finland
- France
- Germany
- Great Britain
- Greece
- Hungary
- Ireland
- Israel
- Italy
- Japan
- Kazakhstan
- Korea
- Malaysia
- Netherlands
- New Zealand
- Norway
- Poland
- Portugal
- Russian Federation
- Saudi Arabia
- Slovak Republic
- Spain
- Sweden
- Switzerland
- South Africa
- Taiwan
- Thailand
- Tunisia
- Turkey
- Ukraine
- UAE
- USA



GELLING MECHANISM OF *High and Low Ester Pectins*

Gelling mechanisms

The association of pectin chains leads to the formation of three-dimensional networks, that is to gel formation.

These are long segments of regular sequence, ruptured by the incorporation of rhamnose and branching of the chain.

Two or more chain segments bond together and start to interact.

High ester pectins jellify predominantly in the so-called “sugar-acid gelling mechanism”, that means a certain amount of acid is required to suppress the dissociation of the free carboxyl groups. In this way, the negative charging of the molecules is prevented and their mutual repulsion reduced.

The degree of esterification determines the type of chain association.

A high sugar concentration lowers the water activity of the system, the pectin molecules are subsequently dehydrated and cluster more easily in bonding zones.

Low ester pectins, too, jellify in a sugar-acid gelling mechanism. Pectin molecules with a low ester content, however, may form bonding zones with bivalent cations, but they may form gels in relative independence from soluble solids content and pH range in the presence of multivalent cations (e.g. calcium ions).

The calcium concentration required for gelling depends here on such parameters as pH of the product, soluble solids content/sugar profile and buffer system.

With smaller additions of calcium, pectin chains will start to cluster over calcium bonds, the viscosity increases. With increased calcium concentration, a gel will form. Overdosage of calcium ions will lead to calcium pectinate precipitation under the given gel formation conditions, this is also called “pre-gelling”.

The gel structure will be less elastic, rather pasty with lower breaking strength. Precipitation of calcium pectinate is only reversible to a limited extent, even when the gel is once more heated above its setting temperature and cooled down without destruction.

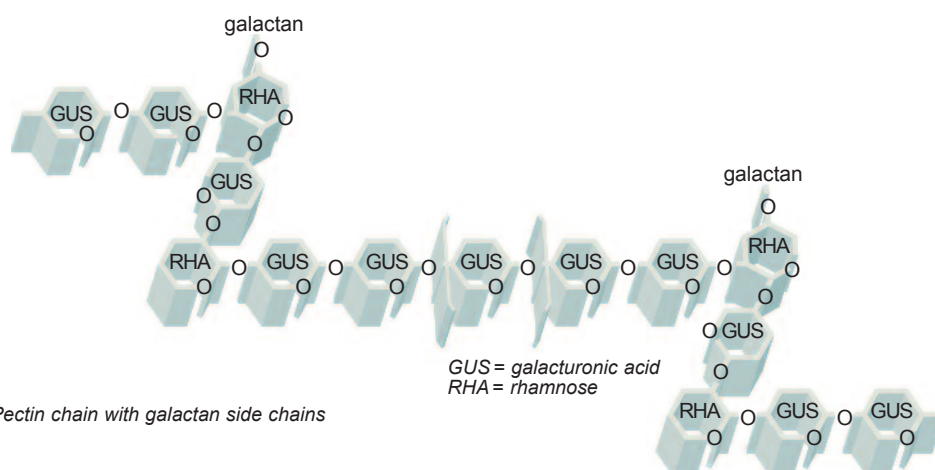


Fig. 3: Pectin chain with galactan side chains

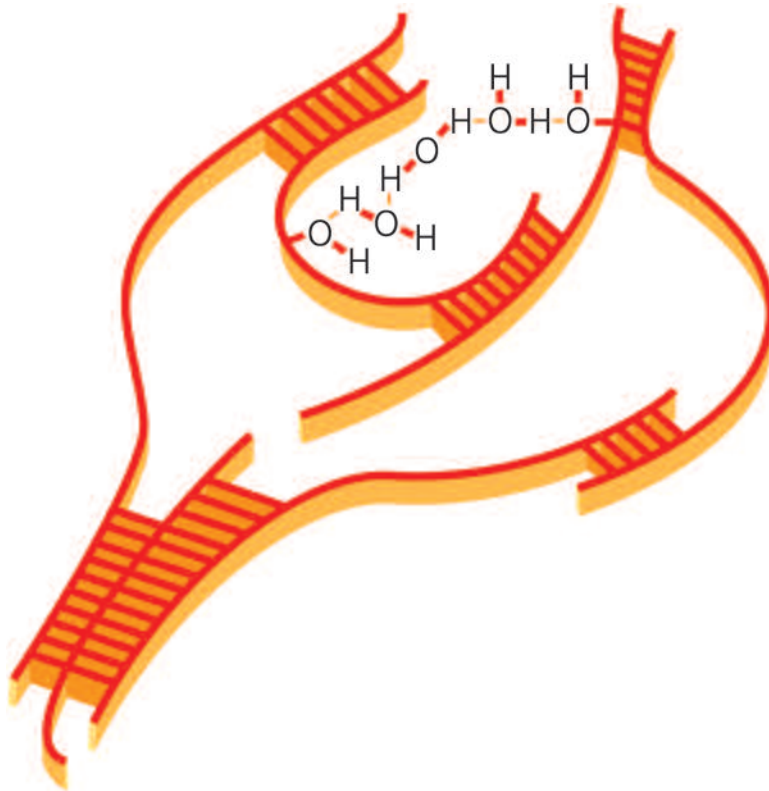


Fig. 4: Gelling mechanism of HM pectins (hydrophobic interaction and hydrogen bonds)

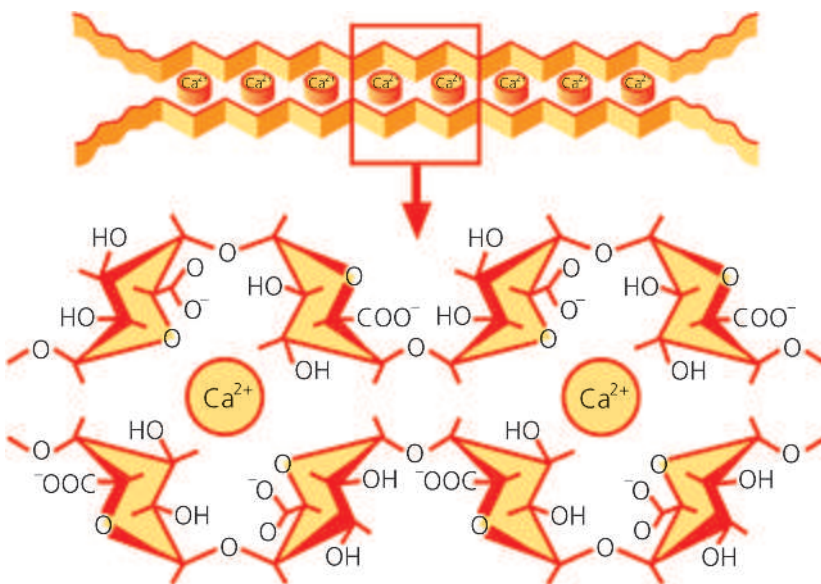


Fig. 5: Gelling mechanism of LM pectins (complexing of calcium ions)

RHEOLOGICAL *Properties*



The application potential of pectin is manifold and textural requirements of users vary greatly; therefore it is one of the most important requirements to develop pectins, which in a targeted and reproducible way reliably control the rheological-sensory characteristics.

Rheological and sensory definitions

In the sensory evaluation of jellified and viscous products we distinguish between the consistency of the gel and the construction of the gel formation, that is the gel structure. To reach an overall impression of the gel, always both criteria have to be jointly considered during sampling.

Our pectins are specifically engineered to control the rheological and sensory properties.

The term **consistency** describes compactness, firmness and viscosity of the sample. These properties are perceived sensorily by pressing and spreading and are generally described as firmness. The behaviour during deformation (pressing) can be rheologically determined, for example, in penetration and compression measuring. Together with the composition of ingredients, such as type of fruit, fruit content and type of sugar, the parameter consistency, all important for sensory acceptance, is also influenced by the choice of pectin.

The term structure or “formation” describes bonding, the gel structure and homogeneity of a sample. The gel structure can be noticed on the surface of a destroyed gel. A rough and brittle structure points to an inhomogeneous gel structure, while a smooth and elastic surface indicates a homogeneous structure.

The ratio of rigid to shifting bonds within a gel formation determines the gel structure. The greater the elastic phases within a gel, the more brittle and fragile the gel structure will be. With an increasing viscous phase, the gel structure will also become increasingly smoother.

Consistency and structure, jointly experienced, will provide the so-called texture. The texture is the overall impression obtained by sensory perception and describes what is experienced in the mouth, the mouth feel, such as softness of bite, the way in which broken pieces disintegrate, tenderness and coating of the tongue during the mastication process.

Texture is the decisive influence on the taste experience.

An important aspect for the sensory evaluation of gels is the release of flavours and aromas during eating. Texture has a decisive influence on the release of these substances. Smooth gels, due to their higher viscous shares, are more aromatic than elastic-brittle gels.



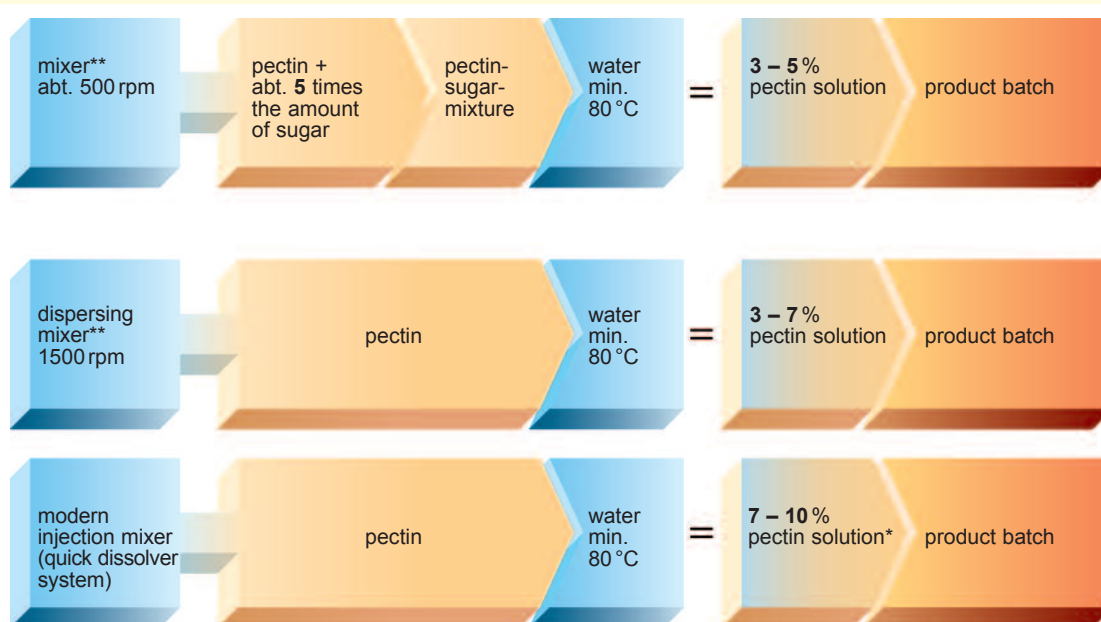
SOLUBILITY/DISPERSIBILITY and Incorporation into the Product Batch

The best way to add standardised pectin during the cooking process is the production of a pectin solution with suitable devices.

However, pectin may also be blended directly into the product batch. In this process, the even and rapid distribution of the pectin powder is of decisive importance.

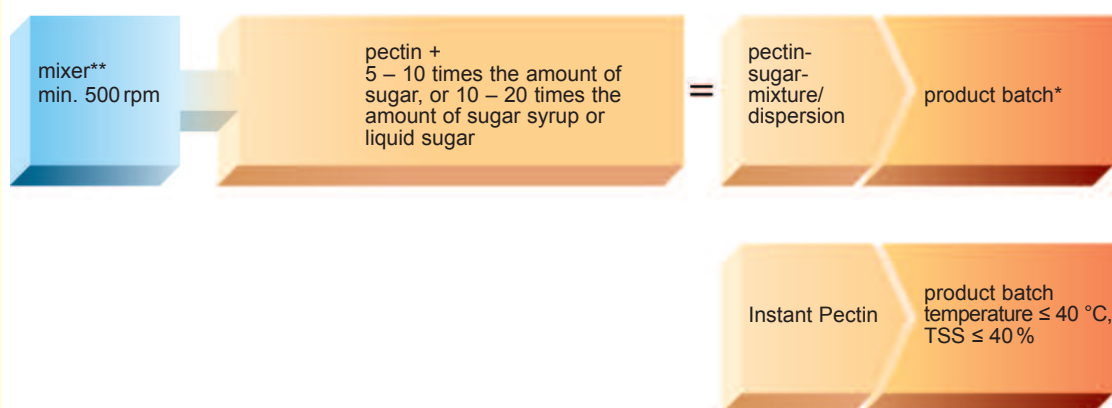
Production of a pectin solution

(The addition of pectin or a pectin sugar premix into the prepared water is done by using a mixer.)



* Benefit: The amount of water to be evaporated during cooking is noticeably smaller

Incorporation of pectin directly into the product batch



* Solids content in batch not more than 30 %.

** Rotation speed depending on dimensioning

Fig. 6: Addition of pectin – as solution or directly to the product batch

CLASSIC, COMBI, AMID AND INSTANT PECTINS – *clearly defined Areas of Application*

We differentiate between the following groups of pectins:

Classic Pectins	= high, medium and low ester apple, citrus, (Betapec) beet, and grapefruit pectins
Amid Pectins	= high and low ester amidated apple and citrus pectins
Combi Pectins	= high ester pectin based on apple and citrus raw material
Instant Pectins	= high, medium and low ester instantised apple and citrus pectins

The pectins within the different groups will form quite typical gel structures, depending on the production process, raw materials and degree of esterification. These are the most important characteristics of the various groups:

Classic Pectins

Our Classic Apple and Citrus Pectins are, due to their consistency, in the majority used in the food industry. Betapec Beet Pectins contain acetyl groups and thus will not gel. They are mostly used for dietary fibre enhancement due to their low viscosity or in pharmaceutical products.

Our pectin experts will ensure the right pectin type is chosen for you.

Amid Pectins

Where flexibility and tolerance towards calcium ions is required, amidated Pectins are recommended, as well as to create specific gel textures in certain products.





Combi Pectins

Combi Pectins with highly specific properties are the result of special extraction conditions. These pectins provide gelling properties somewhat between apple and citrus pectins. The more pronounced viscous properties of apple pectins are supplemented by the brittleness of citrus pectins. The result are pectins which form elastic gels but remain spreadable at the same time and show a relatively low tendency to syneresis.

Instant Pectins

Instant Pectins can provide technological, economic but also qualitative benefits for the user, especially when there is no possibility to use the pectin as solution or pre-dispersion (see page 17).

Instant Pectins are made in a granulation process, in which powdery standardized particles of pectin conglomerate to particles of a bigger size. This effects a porous structure in the individual pectin particles. The space created in this process inside the pectin granulate permits a faster penetration of liquid and – compared to traditional pectins – the wetting of a far larger surface area. This makes lumpfree dispersion in a cold product batch possible and results in a safe solution.

Reliable and quick solubility and shorter cooking times, make for convincing arguments for the use of Instant Pectin.

Three major requirements postulated by the user basically determine what pectin grade or what type of pectin will finally be used:

- product requirements
- formulation
- production technology

H&F PECTINS APPLICATION FOCUS

The Fruit Processing Industry

The appealing properties of jams, marmalades and jellies postulated by the fruit processing industry are to a large degree owing to the benefits of pectin: smooth gel, low syneresis, glossy surface, good spreadability, regular fruit distribution, clear cut and natural, fruit specific flavour.

The incomparable textural properties of Classic Apple Pectins enjoy a worldwide reputation, Classic Citrus Pectins are a meaningful and economic completion of the product range.

The variety of our types of pectin guarantees the realisation of every textural requirement.

Precise balancing of the individual formulation parameters and precise choice of the exactly right Classic, Combi or Amid Pectin will bring about the desired product features. For the fruit processing industry, it is of major importance to have pectins at their disposal which settle fruit pieces regularly already during production and filling and start forming a gel structure at defined temperatures.

Jams, jellies and marmalades are made from fruits or fruit juices, sugar, edible acids and pectin as gelling agent. With a sugar content of more than 58 % and a pH value around 3.0, high ester pectins at a dosage of between 0.35 – 0.50 % are recommended, as these offer the optimal gelling conditions for pectin.

Low ester or LM Amid Pectins are recommended for use in products with a lower sugar content (below 58 %) in some case with an addition of calcium salt.



The structure providing properties as well as enhancing the fruit specific flavour have made the natural product pectin an indispensable substance in the production of jams, jellies and marmalades for many decades now.

Jams, Marmalades, Jellies and Fruit Preparations

Pectin type	DE° [%]	A° [%]	Characteristics and Properties	Main Area of Application
Classic AF 201	72 – 76	–	Apple Pectin, rapid set	jams, marmalades and fruit preparations TSS > 58%, pH 2.8 – 3.2
Classic AF 401	59 – 64	–	Apple Pectin, medium rapid set , smooth gel structure	jams and marmalades filled in glass jars or large size containers fruit preparations and fruit spreads with concentrate TSS > 58%, pH 2.8 – 3.2
Classic AF 501	56 – 63	–	Apple Pectin, slow set , spreadable gel texture	jams, marmalades and jellies TSS > 58%, pH 2.8 – 3.2
Classic AF 504	51 – 58	–	Apple Pectin, optimal fruit distribution even at high filling temperature, smooth gel structure	jams, marmalades and fruit preparations TSS > 58%, pH 2.8 – 3.2
Classic AF 601	48 – 54	–	Apple Pectin, smooth gel structure , spreadable texture	jams, marmalades and fruit preparations TSS > 58%, pH 2.8 – 3.2
Classic AF 605	48 – 54	–	Apple Pectin, very good solubility , even with high soluble solids	gelling agent for household, system 1 + 1
Classic CF 207	> 70	–	Citrus Pectin, rapid set	jams, marmalades and fruit preparations TSS > 58%, pH 2.9 – 3.3
Classic CF 307	65 – 70	–	Citrus Pectin, medium rapid set	jams, marmalades and fruit preparations TSS > 58%, pH 2.9 – 3.3
Classic CF 407	60 – 66	–	Citrus Pectin, slow set	jams, marmalades and fruit preparations TSS > 58%, pH 2.9 – 3.3
Classic CF 501	55 – 61	–	Citrus Pectin, extra slow set	jams, marmalades and fruit preparations TSS > 58%, pH 2.9 – 3.3
Amid AF 005-K	31 – 39	12 – 19	amidated Apple Pectin, low calcium reactivity , anti-floating effect	jams, marmalades and fruit preparations TSS 50 – 60%, pH 3.0 – 3.6
Amid CF 005-B	36 – 42	10 – 16	amidated Citrus Pectin, very low calcium reactivity	jams, marmalades and fruit preparations TSS > 55%, pH 2.8 – 3.2

DE° = Degree of esterification A° = Degree of amidation

Jams, Marmalades, Jellies and Fruit Preparations

Pectin type	DE° [%]	A° [%]	Characteristics and Properties	Main Area of Application
Classic AF 703	38 – 44	–	Apple Pectin, medium calcium reactivity	low-calorie jams and fruit preparations TSS < 55%, pH 3.0 – 3.4 fruit purees TSS 15 – 25 %, pH 3.3 – 3.8 fruit toppings TSS < 55%, pH 3.0 – 3.8
Classic AF 710	29 – 34	–	Apple Pectin, elastic texture, high calcium reactivity	low-calorie fruit preparations TSS 10 – 50 %, pH 3.0 – 3.4 > 40% TSS setting without adding calcium < 40% TSS low calcium addition necessary
Classic CF 701	39 – 45	–	Citrus Pectin, low calcium reactivity	low-calorie jams, fruit spreads and jellies TSS 40 – 60%, pH 3.0 – 3.6
Classic CF 703	32 – 38	–	Citrus Pectin, high calcium reactivity	low-calorie jams, marmalades, fruit spreads and jellies TSS ≤ 45%, pH 3.0 – 3.6

DE° = Degree of esterification A° = Degree of amidation



Jams, Marmalades, Jellies and Fruit Preparations

Pectin type	DE° [%]	A° [%]	Characteristics and Properties	Main Area of Application
Amid AF 005	32 – 40	10 – 16	amidated Apple Pectin, <i>low calcium reactivity</i>	jams, marmalades, low calorie fruit preparations and fruit spreads TSS 50 – 65%, pH 3.0 – 3.5
Amid AF 005-N	27 – 35	12 – 18	amidated Apple Pectin, <i>medium calcium reactivity</i>	calorie-reduced jams and fruit spreads TSS 35 – 50%, pH 3.0 – 3.6
Amid AF 005-Q	24 – 31	12 – 19	amidated apple pectin, <i>medium calcium sensitivity, optimum berry distribution, smooth gel texture</i>	cranberry preparation TSS 35 – 50%, pH 2.8 – 3.3
Amid AF 010	30 – 36	14 – 20	amidated Apple Pectin, <i>medium calcium reactivity</i>	low-calorie jams, marmalades, fruit preparations and fruit spreads TSS 30 – 55%, pH 3.0 – 3.5
Amid AF 020	27 – 32	18 – 23	amidated Apple Pectin, <i>high calcium reactivity</i>	low-calorie jams, marmalades, fruit preparations and fruit spreads TSS 10 – 40%, pH 3.0 – 4.0
Amid CF 005	32 – 40	10 – 16	amidated Citrus Pectin, <i>low calcium reactivity</i>	low-calorie jams, marmalades, fruit preparations and fruit spreads TSS > 45%, pH 3.0 – 3.5
Amid CF 010-D	30 – 36	14 – 20	amidated Citrus Pectin, <i>medium calcium reactivity</i>	low-calorie jams, marmalades, fruit preparations and fruit spreads TSS 30 – 55%, pH 3.0 – 3.5
Amid CF 020-C	27 – 32	18 – 23	amidated Citrus Pectin, <i>high calcium reactivity</i>	low-calorie jams, marmalades, fruit preparations and fruit spreads TSS 15 – 45%, pH 3.0 – 4.0
Amid CF 025-D	19 – 28	22 – 25	amidated Citrus Pectin, <i>very high calcium reactivity</i>	low-calorie jams, marmalades, fruit preparations and fruit spreads TSS 15 – 30%, pH 3.0 – 4.0

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H&F-PECTINS APPLICATION FOCUS

The Baking Industry

Baked Products

Classic Apple Pectins' indispensable characteristics are truly highlighted in baked products, especially in bake stable fruit preparations.

Whether in doughnut fillings, on pies, in gingerbread, on yeast dough or biscuits, in Danish or puff pastry – it is largely owing to pectin that perfect processing can be guaranteed.

High melting temperature and good stability of shape are of paramount importance.

Centre fillings, which are mostly supplied in large-size containers for large-scale production, must show a smooth pumpable, pasty consistency and problem-free metering.

Mechanical stressing, as e. g. is sometimes generated during filling, must not have a negative effect on the gel structure. Furthermore, a high melting temperature and good stability of shape during baking is of highest importance to prevent deformation and leakage of the fruit filling.

The baked product will be of delicious appearance and will keep its typical fruit flavour.

Cake glazings

Cake glazings or “nappage” prevents fruits from drying out and provides cakes and biscuits with a glossy surface. The textural requirements for these products are extremely demanding and may be specifically designed for this application by using standardized Amid Pectins.



Baked Products

1. Baking Stable Fruit Preparations

Pectin type	DE° [%]	A° [%]	Characteristics and Properties	Main Area of Application
Classic AB 401	59 – 64	–	Apple Pectin, baking stable, gelled texture	baking stable fruit preparations (in buckets) TSS ≥ 60%, pH 2.9 – 3.2
Classic AB 901	35 – 42	–	Apple Pectin, very high baking stability pasty, smooth texture	baking stable fruit preparations TSS 40 – 72%, pH 3.0 – 3.8
Classic AB 902	36 – 44	–	Apple Pectin, baking stable, pasty, pumpable texture	baking stable fruit preparations TSS 50 – 72%, pH 2.8 – 3.8
Classic AB 908	31 – 37	–	Apple Pectin, very high baking stability, pasty, pumpable texture	baking stable fruit preparations TSS 50 – 80%, pH 3.0 – 3.8
Amid AB 005	30 – 38	4 – 10	amidated Apple Pectin, high calcium reactivity	baking stable fruit preparations TSS 60 – 80%, pH 3.2 – 6.0

2. Fruit Fillings

Classic AB 902	36 – 44	–	Apple Pectin, pumpable, reversible	fruit fillings for baked products TSS 40 – 72%, pH 3.0 – 3.8
Classic AB 903	37 – 44	–	Apple Pectin, pumpable, reversible, smooth texture	fruit fillings for baked products TSS 40 – 72%, pH 3.0 – 3.8

3. Cake Glazings (Nappage)

Amid CB 025-G	20 – 25	22 – 25	amidated Citrus Pectin, very high calcium reactivity	nappage for dilution 30 % up to 100 % of water TSS 62 – 67 %, pH 3.2 – 3.8 ready to use spray nappage TSS 40 – 45 %, pH 3.3 – 3.8
Amid CB 020-A	27 – 32	18 – 23	amidated Citrus Pectin high calcium reactivity	classic nappage (dilutable with 0 – 30 % water) TSS 62 – 67 %, pH 3.3 – 3.8
Amid CB 005-A	30 – 36	4 – 8	amidated Citrus Pectin low calcium reactivity	cake glazing for cold use TSS 60 – 65 %, pH 3.1 – 3.7

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H&F PECTINS APPLICATION FOCUS

The Confectionery Industry

Producers and consumers of confectionery products have a very clear idea as to what fruit jellies and jelly fillings should taste like. Pectins provide the appealing, elastic structure and enhance naturally the flavour and allow a smooth, brilliant cut.

Our Classic and Amid Pectins promote uninterrupted processing.

The confectionery industry, moreover, greatly appreciates the good solubility and the accurate setting temperature as well as setting time of pectins. Classic and Amid Pectins are standardised to a constant gelling behaviour, dissolve rapidly, are heat stable and thus promote uninterrupted processing.

Pectins have a great application potential in the confectionery field: fruit jellies, fruit gums, gum products with and without gelatine, delicate fruit rolls, tender fillings for chocolates and high boilings, jellies in domino ginger bread, delicious coating, aerated products, jelly layers for bakery products.

Special pectins are used in combination with gelatine to increase the melting temperature.



Confectionery Articles

Jelly Products and Foamed Candies

Pectin type	DE° [%]	A° [%]	Characteristics and Properties	Main Area of Application
Classic AS 501	55 – 61	–	Apple Pectin, slow set , as a rule addition of buffer salts required	fruit jellies, jelly fillings, domino ginger TSS 68 – 80%, pH 2.8 – 3.4
Classic AS 502	56 – 60	–	Apple Pectin, slow set	fruit jellies, jelly fillings, domino ginger TSS 68 – 80%, pH 2.8 – 3.4
Classic AS 507	58 – 65	–	Apple Pectin, medium rapid set , firm, short, elastic texture, smooth break	fruit jellies, jelly fillings, pastilles, gum products TSS 68 – 80%, pH 2.8 – 3.4
Classic CS 501	55 – 61	–	Citrus Pectin, slow set , as a rule addition of buffer salts required	fruit jellies, jelly fillings TSS 68 – 80%, pH 3.0 – 3.6
Classic CS 502	58 – 65	–	Citrus Pectin, medium rapid set , firm, short, elastic texture, smooth break	fruit jellies, jelly fillings, pastilles, gum products TSS 68 – 80%, pH 3.0 – 3.6
Amid CS 005	51 – 59	4 – 9	amidated Citrus Pectin, very slow set , no addition of buffer salts necessary, low acid content possible, low heat viscosity, suitable for high TSS	fruit jellies, jelly fillings, pastilles TSS 68 – 85%, pH 2.8 – 3.6
Amid CS 025-B	16 – 24	20 – 25	amidated Citrus Pectin, high calcium reactivity , suitable for high pH-value	confectionery with increased pH value without addition of acid (e.g. Turkish Delight) TSS 70 – 85%, pH 4.0 – 5.5

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H&F PECTINS APPLICATION FOCUS

The Dairy Industry

Sour milk products

Fruit preparations in dairy products call for pectins, since they provide the required rheological properties and guarantee good dosing, a regular fruit distribution in the container due to their yield point, a homogeneous mixing with the fermented milk product and a good shelf life of the finished product.

In fruit yoghurts, pectins will provide the fruit preparations with a smooth and creamy structure and fruit specific flavour. They also help in a regular distribution of the fruit particles and in achieving a smooth surface. In layered products they have a stabilising effect and keep the fruit preparation separated from the yoghurt.

Pectins for fruit preparations in milk products have been expertly engineered for this application and for specific requirements.

Acidified milk drinks

High ester Classic Citrus and Apple Pectins in yoghurt drinks protect the protein at a low pH range against heat denaturation during the pasteurisation process, thus preventing sedimentation and flocculation. This guarantees a stable product with optimal sensory properties and without loss of quality even over longer storage times.

Yoghurts, milk desserts

Special Classic and Amid Pectins which are able to react with calcium ions in yoghurt improve the texture of yoghurt. This reduces whey syneresis and enhances stability.



Milk-fruit desserts

H&F Classic and Amid Pectins are also used in this area of application. A semi-finished product made of sugar, buffer substance, fruit and water is mixed e.g. with the equal amount of cold milk. The result is a gel which forms minutes after mixing.

Dairy Products

1. Fruit Preparations for Sour Milk Products

Pectin type	DE° [%]	A° [%]	Characteristics and Properties	Main Area of Application
Classic AY 601	46 – 55	–	Apple Pectin, medium calcium reactivity , addition of calcium salts required at low TSS	fruit preparations for yoghurts TSS 30 – 45%, pH 3.2 – 4.0
Classic AY 901	41 – 48	–	Apple Pectin, low calcium reactivity , addition of calcium salts required at low TSS	fruit preparations for yoghurts TSS 45 – 65%, pH 3.5 – 4.2
Amid AY 005-C	34 – 39	6 – 10	amidated Apple Pectin, low calcium reactivity	fruit preparations for yoghurts TSS 25 – 50%, pH 3.5 – 4.5
Classic CY 301	66 – 72	–	Citrus Pectin, increasing viscosity after mixing with yoghurt	fruit preparations for yoghurts TSS 30 – 50%, pH 3.5 – 4.2
Amid CY 025	23 – 28	22 – 25	amidated Citrus Pectin, very high calcium reactivity , high yield point at high temperatures	fruit preparations for yoghurts TSS 25 – 50%, pH 3.5 – 4.2
Amid CF 010	30 – 36	14 – 20	amidated Citrus Pectin, medium calcium reactivity	fruit preparations for yoghurts TSS 30 – 60%, pH 3.5 – 4.5
Amid CF 020	27 – 32	18 – 23	amidated Citrus Pectin, high calcium reactivity	fruit preparations for yoghurts TSS 10 – 40%, pH 3.5 – 4.5

2. Yoghurt Drinks

Classic CM 201	> 68	–	Citrus Pectin, protective colloid	stabilisation of fermented milk drinks pH 3.6 – 4.2 Dosage: 0.3 – 0.5%
Classic CM 203	> 68	–	Citrus Pectin, protective colloid	stabilisation of fermented milk drinks with low viscosity pH 3.6 – 4.2 Dosage: 0.2 – 0.5%

3. Yoghurt, Fruit-Milk Desserts

Classic AM 901	38 – 44	–	Apple Pectin, jellifies with calcium of milk	yoghurt and fresh cheese with creamy structure pH 3.9 – 4.2 fruit-milk desserts jellification by mixing with cold milk TSS 25 – 40%, pH 3.9 – 4.1 semi-finished products for milk-shakes TSS 25 – 40%, pH 3.9 – 4.1
Amid CM 025	23 – 27	23 – 25	amidated Citrus Pectin, very high calcium reactivity, incre- asing viscosity and creaminess, smooth, bright texture, decrease of syneresis	fruit preparations for yoghurts, preferred in fermented products
Amid CM 025-A	31 – 36	18 – 23	amidated Citrus Pectin, high calcium reactivity, increasing firmness, smooth gel, decrease of syneresis	fruit preparations for yoghurts, preferred in fermented products

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H&F PECTINS APPLICATION FOCUS

The Beverage Industry

Juices and soft drinks

Pectins are highly suitable components in soft drinks due to their low-calorie carbohydrate nature and cloud stabilising and viscosity increasing properties.

Classic and Instant Pectins to enhance mouth-feel in low-calorie soft drinks.

Low-calorie soft drinks are becoming ever more popular. They are defined as non-alcoholic beverages in which sugar has been partially or totally substituted by a sweetener or a combination of sweeteners. The resulting loss in full mouthfeel may be improved ideally by the use of Instant Pectins or Classic Pectins.



Beverages and Sherbets

Soft Drinks and Sherbets

Pectin type	DE° [%]	A° [%]	Characteristics and Properties	Main Area of Application
Classic AJ 203	70 – 76	–	Apple Pectin, turbidity stabilizing, viscosity increase	stabilising turbidity in soft drinks pH 2.7 – 4.5 Water ice, sherbets Dosage: 0.05 – 0.30 %
Classic CJ 206	68 – 76	–	Citrus Pectin, turbidity stabilizing, viscosity increase	stabilising turbidity in soft drinks pH 2.7 – 4.5 Water ice, sherbets Dosage: 0.05 – 0.30 %
Instant CJ 204	68 – 72	–	instantized Citrus Pectin, turbidity stabilizing, viscosity increase	stabilising turbidity in soft drinks pH 2.7 – 4.5
Betapec RU 301	> 50	–	sugar beets, pure pectin, unstandardized	emulsion stabilising dietary fibre enrichment

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H&F PECTINS APPLICATION FOCUS

The Delicatessen Sector



Sauces and ketchup

A high-quality tomato ketchup has to meet very strict rheological specifications. The addition of a suitable H&F Pectin will compensate the shortcomings of the native pectin in view to defined yield point and structural viscous behaviour. Furthermore, rheological properties of dips, chutneys and barbecue sauces can be controlled by the use of pectin.

Crème fraîche, mayonnaise and margarine

With the addition of H&F Pectins the water binding capacity and texture of crème fraîche and mayonnaise are improved. Besides a good water binding capacity low-fat margarine shows an excellent spreadability by the addition of H&F Classic Pectin.

Delicatessen

Ketchup, Sauces, Crème fraîche, Mayonnaise, Margarine

Pectin type	DE° [%]	A° [%]	Characteristics and Properties	Main Area of Application
Amid CF 010	30 – 36	14 – 20	amidated Citrus Pectin, medium calcium reactivity	ketchup, sauces and chutneys TSS 20 – 40%, pH 3.0 – 3.8

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H&F PECTINS APPLICATION FOCUS

Pharmaceuticals, Cosmetics and the Non-Food Sector

Pharmaceuticals and cosmetic products

Due to the exploration and the utilisation of the natural properties of pectins, their application tend to become more and more varied and sophisticated. This is convincingly shown in the fields of pharmaceuticals and cosmetics.

The application potential of pectin in the non-food sector is far from being fully exploited.

The styptic and curing effects of natural pectins are fully developed in healing ointments. Furthermore, pectins will help to delay a rise in the glucose blood level and enhance serum cholesterol. Drugs are encapsulated with a pectin film to protect the gastric mucosa and to allow sustained release of the active substance into the blood circulation.

In the cosmetics industry, pectin is being used as a natural structure provider on plant basis for pastes, ointments, oils and creams. In deodorants and tooth pastes, pectin coats special flavour substances, but it is also used as a thickener and stabiliser in hair tonics, body lotions and shampoos.

Non Food products

In the tobacco industry, especially H&F Betapec Beet Pectin is used as a natural adhesive for the wrappers of cigars and cigarillos.

With pectins, ceramic filters with a homogeneous pore-structure can be produced. Even in the production of plastics pectins become more and more important.

This list of very different and varied applications of pectins could be continued at great length. These examples prove the development potential of pectin and the chances and opportunities waiting in the future.



Pharmaceuticals, Cosmetics and Non-Food Products

Dietetic, Pharmaceutical and Cosmetic Products, Dietary Fibre Enrichment

Pectin type	DE° [%]	A° [%]	Characteristic and Properties	Main Area of Application
Classic AU 201 USP	72 – 76	–	Apple Pectin, pure pectin, unstandardized, USP standard	dietetic and pharmaceutical products, e.g. positive effect on serum cholesterol level cosmetic products dietary fibre enrichment
Classic AU 202	68 – 76	–	Apple Pectin, pure pectin, unstandardized	dietetic and pharmaceutical products, e.g. positive effect on serum cholesterol level dietary fibre enrichment
Classic AU 701	38 – 44	–	Apple Pectin, pure pectin, unstandardized	dietetic and pharmaceutical products, e.g. positive effect on - serum cholesterol level - heavy-metal and radionuclide catching cosmetic products
Classic CU 201	70 – 74	–	Citrus Pectin, pure pectin, unstandardized	dietetic and pharmaceutical products cosmetic products
Classic CU 401 USP	60 – 64	–	Citrus Pectin, pure pectin, unstandardized, USP standard	dietetic and pharmaceutical products e.g. positive effect on serum cholesterol level cosmetic products dietary fibre enrichment
Classic CU 701	32 – 38	–	Citrus Pectin, pure pectin, unstandardized	dietetic and pharmaceutical products e.g. positive effect on serum cholesterol level cosmetic products
Classic GU 401 USP	> 60	–	Grapefruit Pectin, pure pectin, unstandardized, USP standard	dietetic and pharmaceutical products e.g. positive effect on serum cholesterol level cosmetic products dietary fibre enrichment
Betapec RU 301	> 50	–	Beet Pectin, pure pectin, unstandardized	dietetic and pharmaceutical products e.g. positive effect on serum cholesterol level cosmetic products dietary fibre enrichment emulsion stabilisation

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INDIVIDUALITY *is our Strength*

Pectins by Herbstreith & Fox have enjoyed a worldwide reputation over many decades. Continual improvement of our production methods and high quality standards have contributed significantly to our success today in the world market. This development has always been characterised by innovative thinking and farsighted research.

Today, we are in a position to offer pectins which can be used in all areas of application that are currently conceivable. Consistent production and quality controls using state-of-the-art analytical equipment guarantee the constant high quality of our pectins.

This positive and continuous progressive approach has not only been upheld in the face of challenges, which our staff in research and development have repeatedly and successfully set themselves, but also by the great variety of requirements brought to us by our customers, the users.

An essential component of this successful cooperation with our users is, of course, the transfer of our know-how. Our analysis can provide valuable assistance even early on in the assessment and control of raw materials.

We also supply users with formulations and production solutions, e.g. for the manufacture of high-quality jams, jellies and marmalades.

For this purpose, our technical experts integrate specific pectins into the composition and optimisation of formulations in order to get the most out of them.

Analysis of your finished product also contributes to assuring the high and consistent quality of your product, and can even shed light on potential improvements regarding your end product.

New, promising product ideas should not fail because of formulation or production problems. This is what we stand for. In the interests of the manufacturer, the product and the consumer.





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